

MATT GRAYLING

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University of Cambridge, UK

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Supernova physics, cosmology, statistics, data science, machine learning

RESEARCH CAREER

Research Associate in Supernovae/Data Science

Oct 2022 – Present

Institute of Astronomy/Kavli Institute for Cosmology, University of Cambridge

- Developing next-generation SED model for fitting type Ia supernova light curves, BayeSN, using latest statistical inference methods.
- Analysing environmental-dependence of intrinsic properties of type Ia supernovae.
- Lead of SN Ia Modelling Topical Team within the Dark Energy Science Collaboration.

Postdoctoral Research Associate

Oct 2024 – Present

Pembroke College, University of Cambridge

- Presenting research and leading small group undergraduate tutorials.

EDUCATION

Astrophysics PhD, CDT in Data-intensive Science

Sep 2018 – Mar 2022

University of Southampton

Supervisor: Prof Mark Sullivan

Awarded March 2023

Thesis: Core-collapse Supernovae in the Dark Energy Survey

Physics MSci

Sep 2014 – Jun 2018

Imperial College London

Supervisor: Dr Dave Clements

Graduated with First Class Honours

Dissertation: Effect of Environment on Galaxy Properties in Intermediate Redshift Clusters

SKILLS

- **Statistical inference and machine learning; jax, numpyro, keras, tensorflow**
Developing cutting-edge, GPU-accelerated hierarchical Bayesian models for fast inference across thousands of parameters; experienced with developing neural networks.
- **Supernova astrophysics and cosmology**
Constraining effect of environment on intrinsic properties of type Ia supernovae.
- **Leadership and organisation**
Leading Type Ia Supernova Modelling Topical Team in Dark Energy Science Collaboration, organise colloquia at Institute of Astronomy as part of colloquium committee.
- **Scientific communication**
Invited speaker at national and international conference, several seminars. Invited reviewer for The Astrophysical Journal, experienced with scientific writing.
- **Collaborative and independent research**
Comfortable working independently and within major international collaborations.

COLLABORATIONS AND LEADERSHIP POSITIONS

Dark Energy Science Collaboration

- Lead of SN Ia Modelling Topical Team, coordinating efforts to develop and compare state-of-the-art type Ia supernova models; organised July 2024 workshop in Stockholm.
- Leading project to incorporate BayeSN within current SN cosmology pipeline for LSST.

Young Supernova Experiment

- 4 observing nights on DECam for follow-up of YSE targets.

SAINTS

- Founding member of newly-established international collaboration compiling and analysing the largest ever sample of type Ia supernovae observed in the near-infrared.

TALKS AND SEMINARS

10 invited seminars and conference talks, including:

The environmental dependence of SN Ia i-band secondary maximum May 2025

Invited Seminar, Duke University, USA

Simulation-based Inference: Connecting SN Cosmology with Astrophysics April 2025

Cosmology and ML Seminar, Center for Computational Astrophysics, USA

Unveiling the Intrinsic SN Ia Mass Step with BayeSN April 2025

Invited Seminar, Rutgers University, USA

Unveiling the Intrinsic SN Ia Mass Step with BayeSN July 2024

European Astronomy Society Annual Meeting, Padova, Italy

Supernovae: An Explosion in Data December 2023

Machine Learning for Transient Science, University of Warwick, UK

Supernovae: An Explosion in Data April 2023

Data Science Seminar Series, SISSA, Trieste, Italy

11 contributed talks across major international conferences and workshops.

GRANTS

Cosmic Lighthouses Conference

Won **£11,000 award** from Kavli Institute to fund Cosmic Lighthouses: Astrophysical and Cosmological Challenges with Type Ia Supernovae in Cambridge.

Type Ia Supernova Modelling Workshop

Won Department of Energy funding to support SN Ia Modelling Workshop, July 2024.

SUPERVISION EXPERIENCE

PhD Second Supervisor: B. Boyd
Oct 2022 – Present

Accounting for Selection Effects in Supernova Cosmology with Simulation-Based Inference

Master's Supervisor: G. Collins
Oct 2025 – Present

Constraining the 3D shape of early galaxies with JWST

Master's Supervisor: A. Reilly
Oct 2024 – Aug 2025

Understanding the reddest type Ia supernovae with hierarchical Bayesian modelling

Master's Supervisor: D. Wentworth
Oct 2024 – Jun 2025

Constraining the 3D shape of early galaxies with JWST

SERVICE

Conference and Workshop Organisation

Cosmic Lighthouses

Co-chair of largest SN Ia conference globally since 2019, with ~80 attendees, in July 2025

Dark Energy Science Collaboration Sprint Week

Local Organising Committee for DESC Sprint Week, Cambridge, Scheduled November 2025

Type Ia Supernova Modelling Workshop

Chair of DESC Type Ia Supernova Modelling Workshop, Stockholm, July 2024

National Astronomy Meeting

Scientific Organising Committee for Exploring the Exploding Transient Diversity with Next-Generation Facilities at NAM, July 2021

Peer Review

Invited Reviewer

Invited reviewer for The Astrophysical Journal

Departmental Roles

Equality, Diversity and Inclusion Committee, 2023 - Present

Member of Equity and Inclusion Working Group, helping organise several themed events for a variety of cultural celebrations

Colloquium Committee, 2023 - Present

Inviting, coordinating with and hosting world-renowned researchers to share their expertise at the University of Cambridge

Teaching Experience

Undergraduate Supervisor, University of Cambridge, 2023 – Present

Part Ia Physics and Stellar Dynamics and Structure of Galaxies, marking undergraduate problems and leading small-group supervisions to discuss problems and course material.

Teaching Assistant, University of Southampton, 2018 – 2022

Physics from Evidence and Computer Techniques in Physics, marking undergraduate assignments and assisting in lab sessions.

Outreach

Public talk, 2024

Cosmic Lighthouses: Charting the Universe with Supernovae, public talk to 200 attendees at departmental open evening at Institute of Astronomy.

Volunteer at outreach event, 2018 – 2020

Volunteered at annual outreach event held at University of Southampton, teaching children about the life cycles of stars and supernovae.

REFEREES

Prof. Kaisey Mandel
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University of Cambridge
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Prof. Mark Sullivan
Associate Dean, Research
University of Southampton
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Prof. Gautham Narayan
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